Year 10 Mathematics

Graphs 2

8.2 Distance-Time Graphs

Distance-Time Graphs

STARTER · 5 MIN

Date: _____ **Title:** Distance-Time Graphs

LEARNING OBJECTIVES

- Read information from a distance-time graph
- Calculate speed from a distance-time graph
- Solve problems involving distance, speed and time

STARTER *Answer in your book. No calculators.*

1. Work out 60 ÷ 2
2. Work out 45 ÷ 1.5
3. How many minutes in 0.5 hours?
4. Work out 25 × 3

Starter — Answers

STARTER

Tick or correct in green pen

1. $60 \div 2 = \mathbf{30}$
2. $45 \div 1.5 = \mathbf{30}$
3. $0.5 \times 60 = \mathbf{30\ minutes}$
4. $25 \times 3 = \mathbf{75}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can read information from a distance-time graph	☐
SILVER	I can calculate speed from a distance-time graph	☐
GOLD	I can solve problems involving distance, speed and time	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Distance-Time Graphs

INTRODUCTION · 3 MIN

A distance-time graph shows how far someone is from a starting point.

Sloping line

The person is moving.

Steeper = faster

Horizontal line

The person has stopped.

Distance stays the same

Speed

The gradient of the line.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

Going back

The line slopes down towards 0.

Returning home

Examples 1 and 2 — Reading a Distance-Time Graph

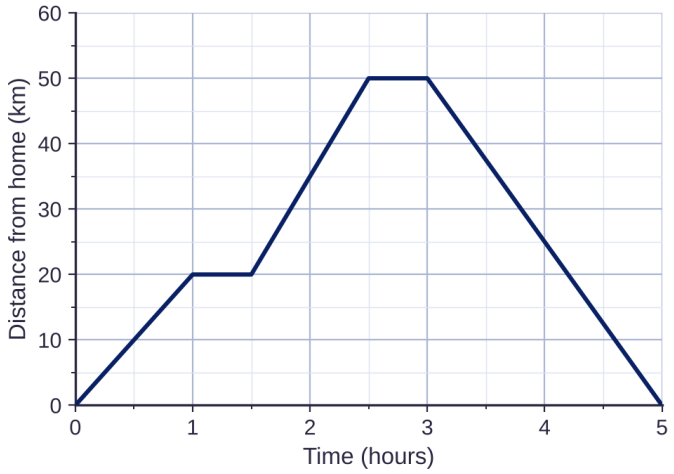
EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

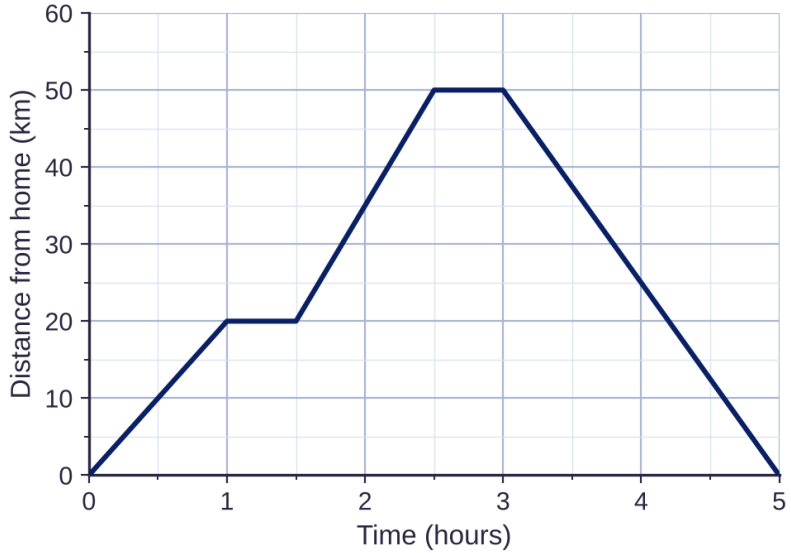
Example 1

Sam cycles from home. How far has he travelled after 1 hour?



Example 2

For how long does Sam stop the first time?



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Read up from 1 hour

Distance **= 20 km**

Tip: read the scale: each small square is 5 km.

Example 2

Horizontal from 1 h to 1.5 h

Stop = 0.5 hours

= 30 minutes

Tip: a flat line means stopped.

Task 1

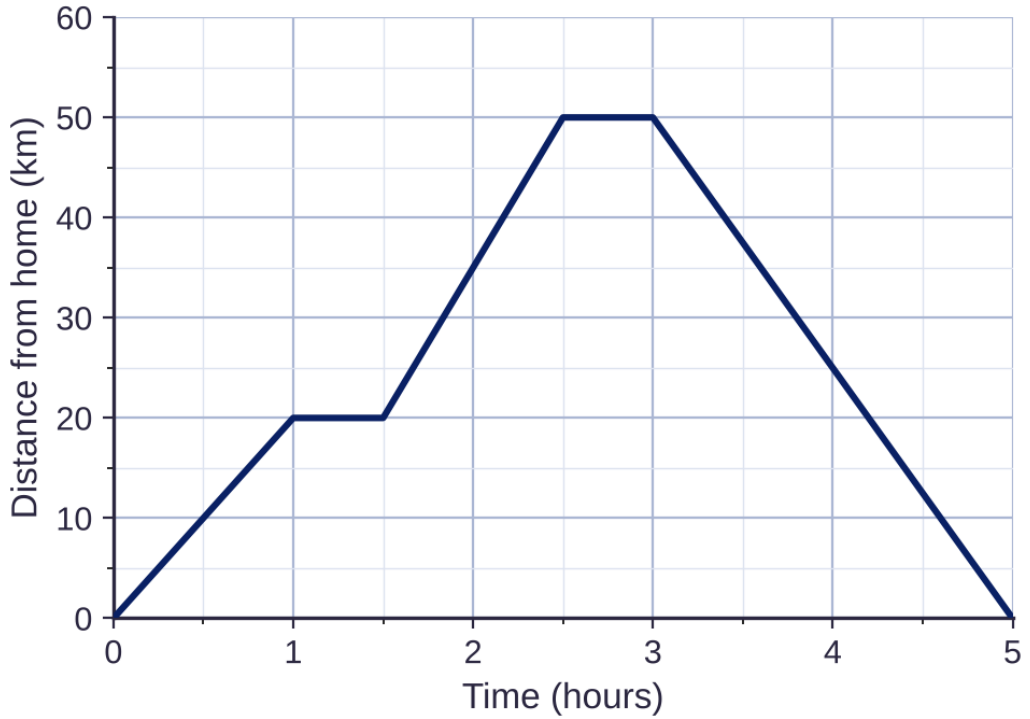
PRACTICE · 9 MIN

BRONZE

Use the graph of Sam's journey

1. How far is Sam from home after 2.5 hours?
2. For how long does Sam stop the second time?
3. How far is Sam's furthest point from home?
4. At what time does Sam get home?
5. What is the total distance Sam cycles?

★ Between which times is Sam travelling fastest?



Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. 2.5 hours = **50 km**
2. 3 − 2.5 = 0.5 hours = **30 minutes**
3. Furthest = **50 km**
4. **Home at 5 hours**
5. 50 + 50 = **100 km**

★ **Between 1.5 h and 2.5 h (steepest line)**

Examples 3 and 4 — Calculating Speed

EXAMPLES · 5 MIN

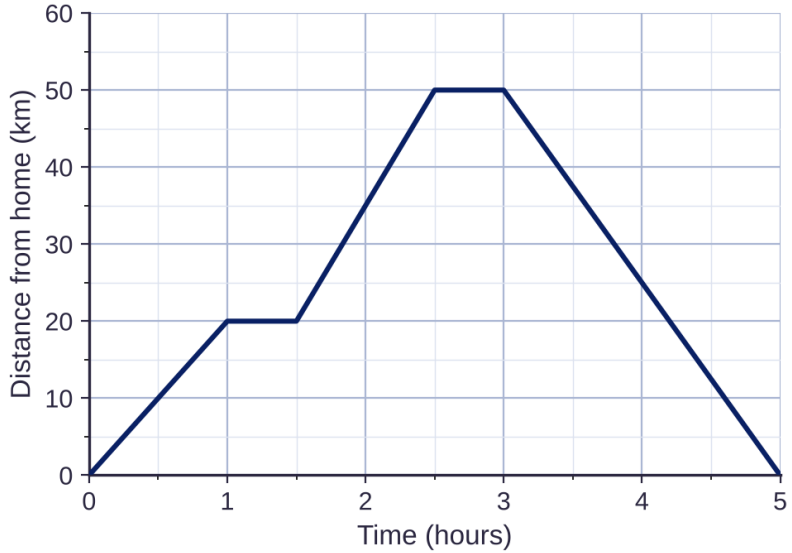
SILVER

Watch and copy into your book

Hint speed = $\frac{\text{distance}}{\text{time}}$

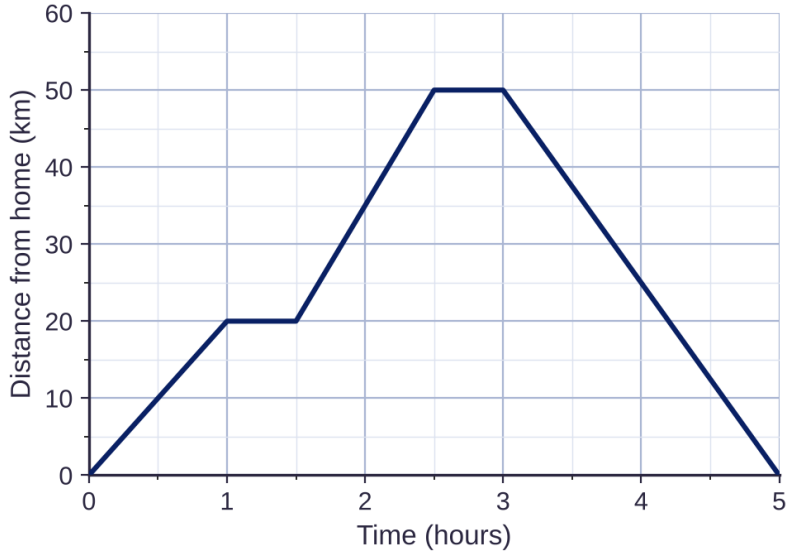
Example 3

Find Sam's speed for the first hour.



Example 4

Find Sam's speed between 1.5 h and 2.5 h.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$\text{Speed} = \frac{20}{1}$$

Speed **= 20 km/h**

Tip: speed = distance ÷ time.

Example 4

Distance = 50 − 20

= 30 km

Time = 1 hour

Speed **= 30 km/h**

Tip: use the change in distance.

Task 2

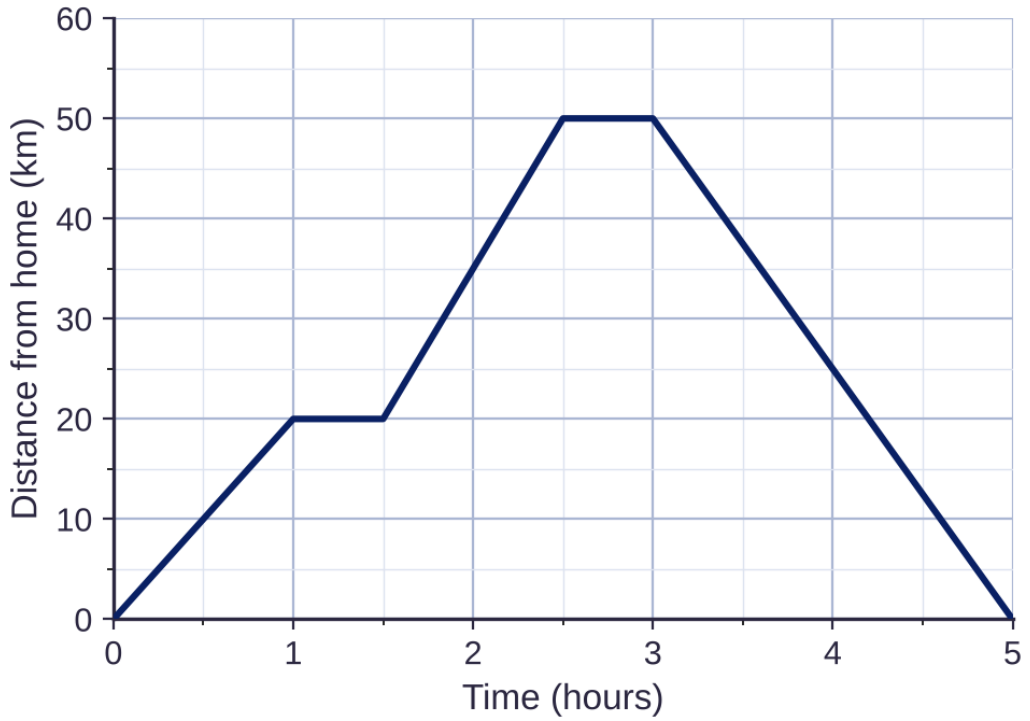
PRACTICE · 9 MIN

SILVER

Use the graph of Sam's journey

Hint speed = $\frac{\text{distance}}{\text{time}}$

1. Find Sam's speed on the journey home.
2. Find the speed of a car that travels 150 km in 2 hours.
3. A runner travels 12 km in 1.5 hours. Find her speed.
4. A train travels at 80 km/h for 2.5 hours. How far does it go?
5. A cyclist travels 45 km at 18 km/h. How long does it take?



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 50 km in 2 hours, $50 \div 2 =$ **25 km/h**

2. $150 \div 2 =$ **75 km/h**

3. $12 \div 1.5 =$ **8 km/h**

4. $80 \times 2.5 =$ **200 km**

5. $45 \div 18 =$ **2.5 hours**

Examples 5 and 6 — Average Speed Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint average speed = $\frac{\text{total distance}}{\text{total time}}$

Example 5

(2 marks)

Find Sam's average speed for the whole journey,

including stops.

Example 6

(3 marks)

Amira drives 120 km in 1 h 30 min.

Find her average speed.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Total distance = 100 km

Total time = 5 hours

Average speed **= 20 km/h**

Tip: average speed = total distance ÷ total time.

Example 6

1 h 30 min = 1.5 h

120 ÷ 1.5

Speed **= 80 km/h**

Tip: change minutes into a decimal of an hour.

Task 3

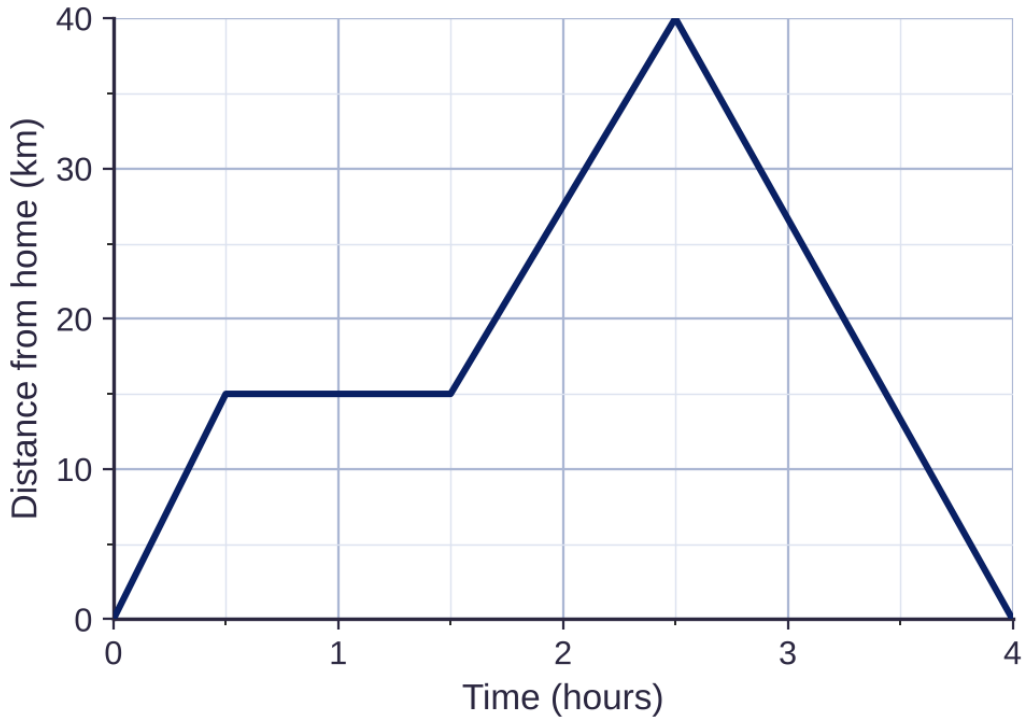
PRACTICE · 9 MIN

GOLD

Questions 1–3 use the graph of Lina's journey

Hint average speed = $\frac{\text{total distance}}{\text{total time}}$

1. Find Lina's speed for the first 30 minutes. (2 marks)
2. Find Lina's speed on the way home. (2 marks)
3. Find Lina's average speed for the whole journey. (2 marks)
4. A bus travels 90 km in 1 hour 15 minutes. Find its average speed. (3 marks)
5. A car travels at 72 km/h for 25 minutes. How far does it go? (3 marks)



Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $15 \div 0.5 = \mathbf{30\ km/h}$

2. $40 \div 1.5 = \mathbf{26.7\ km/h}$

3. $80 \div 4 = \mathbf{20\ km/h}$

4. $1\text{ h }15\text{ min} = 1.25\text{ h}, \quad 90 \div 1.25 = \mathbf{72\ km/h}$

5. $72 \times \frac{25}{60} = \mathbf{30}\text{ km}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE What does a horizontal line on a distance-time graph mean?

SILVER A car travels 90 km in 1.5 hours. Find its speed.

GOLD Find the average speed for 100 km in 2 h 30 min.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **The person has stopped**

SILVER $90 \div 1.5 =$ **60 km/h**

GOLD $2\text{ h }30\text{ min} = 2.5\text{ h}, \quad 100 \div 2.5 =$ **40 km/h**

Exam Practice — Reading Distance-Time Graphs

EXAM PRACTICE · 10 MIN

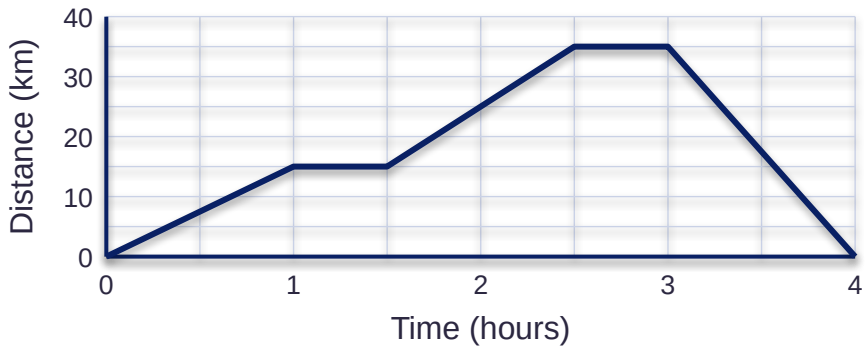
BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

The graph shows Priya's bike ride.



(a) How far has she gone at 1 h? (1)

..... km

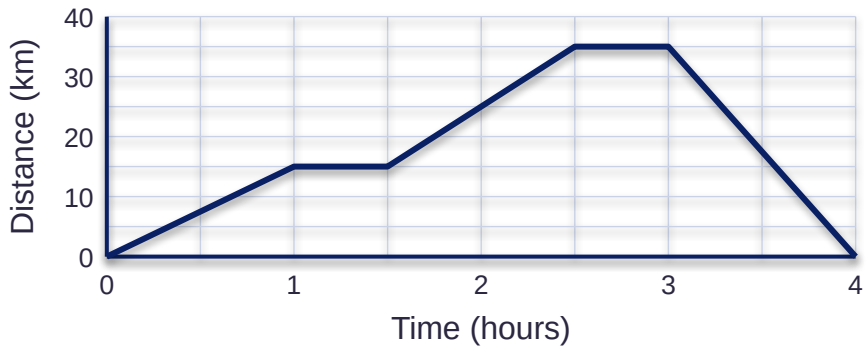
(b) How long is her first stop? (1)

..... min

Question 2

(2 marks)

The graph shows Priya's bike ride.



(a) What is her greatest distance? (1)

..... km

(b) Describe her ride after 3 h. (1)

.....

Exam Practice — Speed from a Graph

EXAM PRACTICE · 10 MIN

SILVER

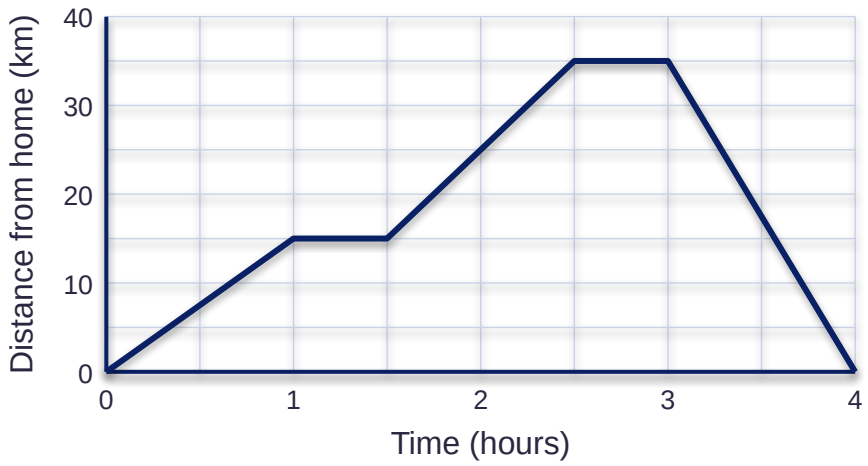
IGCSE exam-style: show all your working

Hint speed = $\frac{\text{distance}}{\text{time}}$

Question 3

(2 marks)

Work out Priya's speed in the first hour.

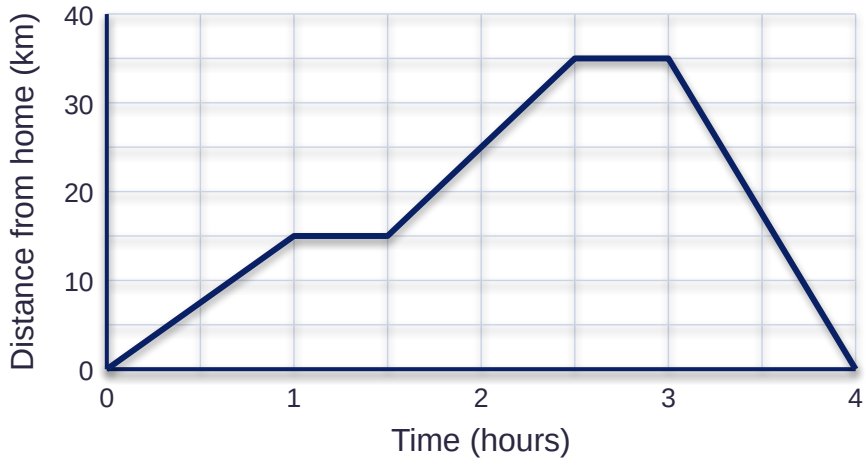


..... km/h

Question 4

(2 marks)

Work out Priya's speed between 1.5 h and 2.5 h.



..... km/h

Exam Practice — Average Speed Problems

EXAM PRACTICE · 10 MIN

GOLD

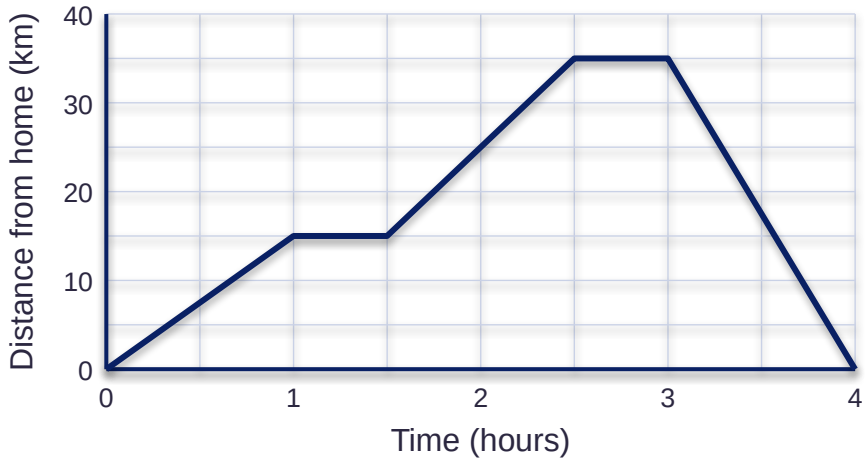
IGCSE exam-style: show all your working

Hint average speed = $\frac{\text{total distance}}{\text{total time}}$

Question 5

(2 marks)

Work out Priya's average speed for the whole journey, including stops.



..... km/h

Question 6

(3 marks)

A train travels 210 km at an average speed of 84 km/h.

It leaves at 09:40.

At what time does it arrive?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 15 km B1

(b) 30 minutes B1

Tip: a flat line means stopped.

Question 2

(a) 35 km B1

(b) She rides back home B1

Tip: a line going down means returning.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

15 ÷ 1 — M1

= 15 km/h — A1

Tip: speed = distance ÷ time.

Question 4

(35 − 15) ÷ (2.5 − 1.5) — M1

= 20 km/h — A1

Tip: speed = gradient of the line.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

35 + 35 = 70 km — M1

70 ÷ 4 = 17.5 km/h — A1

Tip: total distance ÷ total time.

Question 6

210 ÷ 84 = 2.5 h — M1

= 2 h 30 min — M1

12:10 — A1

Tip: 0.5 h = 30 minutes, not 50.